Order-five trace kernel: Lean proof boundary for FROB C5

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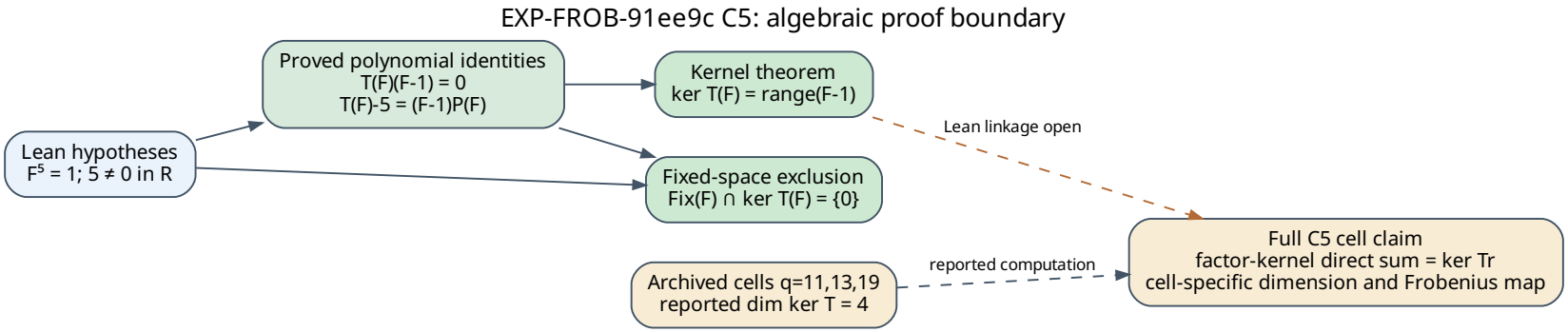
Proved proposition. Over any field R and R-vector space V, let F be a linear endomorphism with $F^5 = 1$ and $5 \neq 0$ in R. For $T(F) = 1 + F + F^2 + F^3 + F^4$, Lean proves $\ker T(F) = \mathrm{range}(F - 1)$ and $\mathrm{Fix}(F) \cap \ker T(F) = \{0\}$. The proof source is `formal/CryptoResearch/FrobTrace/OrderFive.lean`; the frozen target is `formal/targets/frob-c5-order-five-trace-kernel.yaml`.



Boundary. The theorem does not construct the archived $q = 11, 13, 19$ finite fields, identify F with their Frobenius maps, prove the reported dimension four, or prove the nontrivial factor-kernel direct sum. Those are separate C5 obligations. C8's independent control-arm recheck also remains open. No performance or attack-cost claim follows from this lemma.

Sources. EXP-FROB-91ee9c `specification.yaml` control C5; `analysis.md` Section 0; frozen `TASK-20260914-7119f2/report.md`; 2026-10-22 provenance audit.